CAS CS 538. Solutions to Problem Set 9

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How to submit: See instructions on Problem Set 1.

An Efficient DL-based PRF

Do not be intimidated by the long problem. It's broken down into four small parts and each part is quite doable. At the end of the four parts, you get a pretty cool theorem.

The goal of the problem below is to show the following. Let $\mathbb{G}$ be a group of prime order q generated by an element g. Let k be some constant. Let $\alpha_0, \ldots, \alpha_\ell$ be elements of $\mathbb{G}$. For an input x that consists of n bits $b_1, \ldots, b_\ell$, let $F((\alpha_0, \ldots, \alpha_\ell), x)$ be computed as follows:

1. Compute
$$\alpha_x = \alpha_0 \cdot \left(\prod_{i \text{ such that } b_i = 1} \alpha_i \right) = \alpha_0 \cdot \left(\prod_{i=1}^{\ell} \alpha_i^{b_i} \right)$$
2. Output g_x^α

You will prove that $F((\alpha_0, \ldots, \alpha_\ell), x)$ is a PRF, where $(\alpha_0, \ldots, \alpha_\ell)$ is the key and x is the input.

The big idea of the proof is the same as the Goldreich-Goldwasser-Micali binary tree construction of Section 4.6: this PRF can be viewed as a binary tree of depth ℓ, and each level is a PRG. Here is how. Let $h = g_0^\alpha$.

- The root level contains one node h.
- The next level contains two nodes h and h^{α_1}.
- The next level contains four nodes $h, h^{\alpha_2}, h^{\alpha_1}, h^{\alpha_1\alpha_2}$
- The next level contains eight nodes $h, h^{\alpha_3}, h^{\alpha_2}, h^{\alpha_2\alpha_3}, h^{\alpha_1}, h^{\alpha_1\alpha_3}, h^{\alpha_1\alpha_2}, h^{\alpha_1\alpha_2\alpha_3}$.
- ...
- The level at depth d contains g^{α_x} for every binary string x where the first d bits are arbitrary and the last $\ell - d$ bits are 0.
- ...
- The last level contains g^{α_x} for every for every binary string of length ℓ

All we need to prove is that each level d is a secure PRG when the corresponding α_d is kept secret. Thus, let $J : \mathbb{G}^k \times \mathbb{Z}_q \to \mathbb{G}^{2k}$ be defined as follows: on input $(X_1, \ldots, X_k, \alpha)$, compute $Y_i = X_i^\alpha$ for $i = 1 \ldots k$. Output $(X_1, Y_1, X_2, Y_2, \ldots, X_k, Y_k)$.

Theorem 1. *Under the DDH assumption (Definition 10.8 in Section 10.5) on $\mathbb{G}$, J is a secure pseudorandom generator.*

You will now prove this theorem by proving the following parts.

Problem 1. (60 points at 15 each)

(a) Consider the following assumption we call DDH$'$. It is exactly the same as DDH described in Attack Game 10.6 and Definition 10.8, except in Experiment 1, γ is chosen at random from $\mathbb{Z}_q - \{\alpha\beta\}$ (call this modified experiment $1'$ and the corresponding advantage DDH$'\mathsf{adv}$). Show that assuming DDH holds for $\mathbb{G}$, then so does DDH$'$. (*Hint:* apply the difference lemma to prove that experiments 1 and 1' are indistinguishable.)

Solution. Let $\mathcal{A}$ be some adversary. Let W_0, W_1, and $W_{1'}$ be the events that $\mathcal{A}$ outputs 1 in Experiment 0, Experiment 1, and Experiment $1'$ of the respective games. Then, by definition:

$$\mathrm{DDH}\mathsf{adv}[\mathcal{A}] = |\Pr[W_0] - \Pr[W_1]| \quad \text{and} \quad \mathrm{DDH}'\mathsf{adv}[\mathcal{A}] = |\Pr[W_0] - \Pr[W_{1'}]|\,.$$

Now, we can couple Exp 1 and Exp $1'$ on the same underlying probability space. In both, the challenger will sample $\alpha, \beta \xleftarrow{\mathrm{R}} \mathbb{Z}_q$, and then compute $u = g^\alpha, v = g^\beta$. In Exp 1 specifically, $\gamma \xleftarrow{\mathrm{R}} \mathbb{Z}_q$. In Experiment $1'$, $\gamma \xleftarrow{\mathrm{R}} \mathbb{Z}_q \setminus \{\alpha\beta\}$. Now, let's say that the "bad event", Z, is when $\gamma = \alpha\beta$ in Exp 1. Note that γ is uniform in $\mathbb{Z}_q$ and is independent to α, β, $\Pr[Z] = 1/q$. Conditioned on $\neg Z$, the distributions in both experiments are identical. Thusly, by the difference lemma:

$$|\Pr[W_1] - \Pr[W_{1'}]| \leq \Pr[Z] = \frac{1}{q}.$$

And by the triangle inequality:

$$\mathrm{DDH}'\mathsf{adv}[\mathcal{A}] = |\Pr[W_0] - \Pr[W_{1'}]| \leq |\Pr[W_0] - \Pr[W_1]| + |\Pr[W_1] - \Pr[W_{1'}]| \leq \mathrm{DDH}\mathsf{adv}[\mathcal{A}] + \frac{1}{q}.$$

Assuming DDH holds for $\mathbb{G}$, then DDH$\mathsf{adv}[\mathcal{A}]$ is negligible. Since q is exponential in the security parameter, $1/q$ is then also negligible. Thus, DDH$'\mathsf{adv}[\mathcal{A}]$ is negligible, proving DDH$'$ holds for $\mathbb{G}$.

(b) Prove the following.

Lemma 1. Fix any α, β, and $\gamma \not\equiv_q \alpha\beta$ in $\mathbb{Z}_q$. Define the map $M : \mathbb{Z}_q \times \mathbb{Z}_q \to \mathbb{Z}_q \times \mathbb{Z}_q$ as follows. Given inputs d, e:

$$\begin{aligned} x &= (d + \beta e) \bmod q\,; \\ y &= (\alpha d + \gamma e) \bmod q\,; \\ &\text{Output } (x, y)\,. \end{aligned}$$

This map is a bijection.

Hint: Find the inverse of M by solving a system of two linear equations. Linear algebra extends to modular arithmetic modulo a prime q. Note that α, β, and γ are not inputs to the map; rather, they are used to specify a particular map.

Solution. Note the linear system mod, q, that's given:

$$\begin{cases} x \equiv d + \beta e \pmod{q}, \\ y \equiv \alpha d + \gamma e \pmod{q}. \end{cases}$$

Now, look at the first congruence. Solving for d gets us:

$$d \equiv x - \beta e \pmod{q}.$$

And by subbing this into the 2nd congruence:

$$y \equiv \alpha(x - \beta e) + \gamma e \equiv \alpha x + (\gamma - \alpha\beta)e \pmod{q}.$$

Rearrange the terms:

$$(\gamma - \alpha\beta)e \equiv y - \alpha x \pmod{q}.$$

And since $\gamma \not\equiv_q \alpha\beta$ and q is prime, $\gamma - \alpha\beta \not\equiv 0 \pmod{q}$. In the field $\mathbb{Z}_q$, every non-0 element has a multiplicative inverse, so it is possible to solve uniquely for e:

$$e \equiv (\gamma - \alpha\beta)^{-1}(y - \alpha x) \pmod{q}.$$

And by subbing this expression for e back into $d \equiv x - \beta e$, this gives us a unique value for d mod q. Thus, for every pair $(x, y) \in \mathbb{Z}_q \times \mathbb{Z}_q$, there exists exactly one pair (d, e) that maps to it under M. This makes an explicit inverse map M^{-1}, proving that M is invertible. And since M is a map on a finite set and is invertible, it is a bijection.

(c) Prove the following. (*Hint:* work in the exponent space—i.e., consider logarithms of X and Y base g — and use the previous part. The variable naming here deliberately works well with the previous part.)

Lemma 2. For any $u = g^\alpha$, $v = g^\beta$, and $w_1 \neq g^{\alpha\beta}$, if d and e are a uniformly random pair of elements from $\mathbb{Z}_q$, then $X = g^d v^e$ and $Y = u^d w_1^e$ is a uniformly random pair of elements from $\mathbb{G}$ (i.e., every pair X, Y occurs with probability $1/q^2$).

Solution. Let $\gamma = \log_g(w_1)$. Now, since $w_1 \neq g^{\alpha\beta}$, then we have $\gamma \not\equiv_q \alpha\beta$. This is because g generates $\mathbb{G}$ of prime order q, which is the discrete log map $z \mapsto \log_g(z)$, which is a bijection from $\mathbb{G}$ to $\mathbb{Z}_q$.

Now let $x = \log_g(X)$, and $y = \log_g(Y)$. By subbing in $u = g^\alpha$, $v = g^\beta$, and $w_1 = g^\gamma$ into the definitions of X and Y, we get the following system mod q:

$$\begin{cases} x \equiv d + \beta e \pmod{q}, \\ y \equiv \alpha d + \gamma e \pmod{q}. \end{cases}$$

This is exactly the map $M : \mathbb{Z}_q \times \mathbb{Z}_q \to \mathbb{Z}_q \times \mathbb{Z}_q$ from Lemma 1 (1b). And since $\gamma \not\equiv_q \alpha\beta$, (1b) guarantees that M is a bijection.

Because (d, e) is chosen uniformly at random from $\mathbb{Z}_q \times \mathbb{Z}_q$ and M is a bijection, the image $(x, y) = M(d, e)$ must then also be uniformly distributed over $\mathbb{Z}_q \times \mathbb{Z}_q$. By applying the bijection $z \mapsto g^z$ component-wise, it is then the case that $(X, Y) = (g^x, g^y)$ is uniformly distributed over $\mathbb{G} \times \mathbb{G}$. Therefore, every pair $(X, Y) \in \mathbb{G}^2$ occurs with probability exactly $1/q^2$.

(d) Use the previous part to build a reduction that shows DDH′ implies the security of J. (Combined with part (a), this proves Theorem 1.) In particular, show that for every efficient adversary $\mathcal{A}$ breaking J with advantage ϵ, there is an efficient adversary $\mathcal{A}'$ breaking DDH′ with advantage ϵ as well. (*Hint:* This is a single reduction; no hybrid arguments are needed. The previous part tells you exactly how your reduction should transform u, v, and w that it receives from the DDH′ challenger. It should that transformation k times.)

Solution.

Let $\mathcal{A}$ be the adv. that breaks the PRG J with advantage ϵ. Then, we can make another adv. $\mathcal{A}'$ that breaks the DDH′ assumption with the same advantage.

$\mathcal{A}'$ will get a challenge triple (u, v, w) from the DDH′ challenger. $\mathcal{A}'$ will then do the following:

1. $\mathcal{A}'$ takes $2k$ independent values $d_1, e_1, \ldots, d_k, e_k \xleftarrow{\text{R}} \mathbb{Z}_q$.
2. For each $i \in \{1, \ldots, k\}$, $\mathcal{A}'$ will calculate:
$$X_i = g^{d_i} v^{e_i} \quad \text{and} \quad Y_i = u^{d_i} w^{e_i}.$$
3. $\mathcal{A}'$ then sends the sequence $(X_1, Y_1, \ldots, X_k, Y_k)$ to $\mathcal{A}$.
4. $\mathcal{A}'$ will output whatever $\mathcal{A}$ outputs.

Now, by analyzing the distribution of the sequence passed to $\mathcal{A}$ in the two Exp's of the DDH′ game:

- **Exp 0 of DDH′:** Here, $u = g^\alpha$, $v = g^\beta$, and $w = g^{\alpha\beta}$ for some rnd $\alpha, \beta \xleftarrow{\text{R}} \mathbb{Z}_q$.
  For each i, we have:
$$Y_i = (g^\alpha)^{d_i} (g^{\alpha\beta})^{e_i} = g^{\alpha(d_i + \beta e_i)} = (g^{d_i + \beta e_i})^\alpha = X_i^\alpha.$$
  Now, since d_i, e_i are rnd and independent, $X_i = g^{d_i}(g^\beta)^{e_i}$ is randomly distributed in $\mathbb{G}$. Thus, the sequence $(X_1, X_1^\alpha, \ldots, X_k, X_k^\alpha)$ is distributed exactly as the output of J on a uniformly random seed $(X_1, \ldots, X_k, \alpha)$. This is in line with **Exp 0** of the PRG security game.
- **Exp 1′ of DDH′:** Here, $u = g^\alpha$, $v = g^\beta$, and $w = g^\gamma$ where $\gamma \not\equiv_q \alpha\beta$. By Lemma 2 (1c), for fixed u, v, w satisfying this condition, the pair $(X_i, Y_i) = (g^{d_i} v^{e_i}, u^{d_i} w^{e_i})$ is uniformly distributed over $\mathbb{G} \times \mathbb{G}$ when d_i, e_i are rnd. Since the pairs are rnd, the entire sequence $(X_1, Y_1, \ldots, X_k, Y_k)$ is uniformly distributed over $\mathbb{G}^{2k}$. This is in line with **Exp 1** of the PRG security game.

Now since $\mathcal{A}'$ perfectly simulates the PRG game for $\mathcal{A}$, then the advantage of $\mathcal{A}'$ is exactly equal to the advantage of $\mathcal{A}$:

$$\mathrm{DDH}'\mathsf{adv}[\mathcal{A}'] = \mathrm{PRG}\mathsf{adv}[\mathcal{A}, J] = \epsilon.$$

Thus, if DDH′ holds, J is a secure PRG. And by (1a), this proves that J is secure under the standard DDH assumption.

Backdoored PRGs

Problem 2. (40 points at 20 points each)

In 2006, the National Institute for Standards and Technology (NIST) unveiled a new standard called Dual_EC_DRBG, a PRG construction designed by the NSA. The construction was peculiar. It was clunkier than other existing PRGs, and up to 1000 times slower. Then, in 2007, two Microsoft cryptographers discovered something very unusual: the scheme had the possibility to contain a backdoor, which would allow an adversary to predict all future randomness given some output of the PRG. The existence of such a backdoor was confirmed by the Snowden disclosures in 2013: the NSA had indeed inserted a backdoor into this PRG.

Dual_EC_DRBG is designed as a stateful PRG. At its core is a PRG G which is repeatedly composed with itself sequentially, as in section 3.4.2 of the textbook. We know that this construction is secure as long as the underlying G is a secure PRG.

The following is a simplified version of the Dual_EC_DRBG PRG[1]. Let p be a Sophie Germain prime, and $q = (p-1)/2$. Let g and h be generators of QR_p, sampled independently and uniformly. Let $\mathrm{abs} : QR_p \to \mathbb{Z}_q$ be the bijection defined in 8.5.1, such that $\mathrm{abs}(z)$ returns z if $z < p/2$ and $-z$ otherwise. The underlying PRG is as follows:

$$\boxed{\begin{array}{l} G(s_0): \\ \quad r \leftarrow h^{s_0} \\ \quad s_1 \leftarrow \mathrm{abs}\left(g^{s_0}\right) \\ \quad \text{return } (r, s_1) \end{array}}$$

(a) Prove that when g and h are sampled uniformly at random, G is a secure PRG, assuming DDH is hard in QR_p. To capture the notion of g and h being generated uniformly at random, we need to add a step to the PRG security game (Attack Game 3.1 in the textbook): we will have the challenger sample g and h and send them to the adversary as the first step, and then use those values when computing G in Experiment 0.

Solution. Assume, for the sake of contradiction, $\mathcal{A}$ is an eff. adv. that can distinguish G with non-negl. advantage ϵ. Then, it is possible to make another eff. adv. $\mathcal{B}$ that breaks QR_p with the same advantage.

$\mathcal{B}$ will get (u, v, w) from the DDH challenger, where $u = g^\alpha$, $v = g^\beta$, and w is either $g^{\alpha\beta}$ (Exp 0) or g^γ for rnd γ (Exp 1). $\mathcal{B}$ will then do the following:

[1] The real scheme uses an elliptic curve group instead of QR_p, and a different bijection is used. It also removes some bits of the PRG output. What this problem calls g and h are referred to as Q and P.

1. $\mathcal{B}$ sends (u, v) to $\mathcal{A}$ as the public parameters (g, h).
2. $\mathcal{B}$ will calculate $r \leftarrow w$ and $s_1 \leftarrow \text{abs}(u)$.
3. $\mathcal{B}$ will then send the pair (r, s_1) to $\mathcal{A}$.
4. $\mathcal{B}$ outputs whatever $\mathcal{A}$ outputs.

Now look at the distribution of (r, s_1) passed to $\mathcal{A}$ in the two DDH experiments:

- **Exp 0 (Real):** Here, $w = g^{\alpha\beta}$. Let the implicit PRG seed be $s_0 = \alpha$. Then:

  $$r = w = g^{\alpha\beta} = (g^{\beta})^{\alpha} = v^{\alpha} = h^{s_0},$$

  and

  $$s_1 = \text{abs}(u) = \text{abs}(g^{\alpha}) = \text{abs}(g^{s_0}).$$

  This is just the same output distribution as $G(s_0)$ in **Exp 0** of the PRG game.

- **Exp 1 (Rnd):** Here, $w = g^{\gamma}$ where γ is chosen randly and independently of α, β. Thus, $r = w$ is uniformly distributed in QR_p. Since α is uniform, $u = g^{\alpha}$ is also uniformly distributed in QR_p.

  Because $abs : QR_p \to Z_q$ is a bijection (from question, 8.5.1), $s_1 = abs(u)$ is uniformly distributed in Z_q. Moreover, r and s_1 are independent because γ and α are independent. Then, this just becomes the same uniform distribution as in **Exp 1** of the PRG game.

Now since $\mathcal{B}$ perfectly simulates the modified PRG security game for $\mathcal{A}$, the advantage of $\mathcal{B}$ just becomes that of $\mathcal{A}$:

$$\text{DDH}\mathsf{adv}[\mathcal{B}] = \text{PRG}\mathsf{adv}[\mathcal{A}] = \epsilon.$$

By our initial assumption, DDH is hard in QR_p, so $\text{DDH}\mathsf{adv}[\mathcal{B}]$ must be negligible. Therefore, $\text{PRG}\mathsf{adv}[\mathcal{A}]$ is negligible, proving that G is a secure PRG.

(b) Now show that if g and h are generated maliciously, the scheme has a backdoor. In particular, describe how to generate g and h in a way that produces a trapdoor key sk and an efficient function T that takes in sk and r_i (the i-th output of G using g and h, using an unknown random seed), and computes the next state s_{i+1}. (Note that once you know the state, you can compute all future outputs of the PRG.)

Your maliciously generated g and h should be indistinguishable from an honestly generated g and h. This ensures that the existence of an intentional backdoor is undetectable.

Solution. The NSA generator will do the following:

1. Sample $g \xleftarrow{\text{R}} QR_p$ uniformly at rnd.
2. Sample $c \xleftarrow{\text{R}} \mathbb{Z}_q$ uniformly at rnd.
3. Set $h = g^c$.
4. Output the public parameters (g, h), while keeping the trapdoor key $sk = c$ secret.

Now, given the trapdoor $sk = c$ and the i-th output $r_i = h^{s_i}$, the backdoor function T should recover the next state s_{i+1} as follows:

$$T(sk, r_i) = \text{abs}\left(r_i^{c^{-1} \bmod q}\right).$$

Correctness: Since $r_i = h^{s_i} = (g^c)^{s_i} = g^{cs_i}$, then raising r_i to the power of $c^{-1} \bmod q$ will result in:

$$r_i^{c^{-1}} = \left(g^{cs_i}\right)^{c^{-1}} = g^{s_i}.$$

By applying the bijection $\text{abs}(\cdot)$ to g^{s_i}, one will recover $s_{i+1} = \text{abs}(g^{s_i})$. Thus, by knowing sk, one should be able to efficently predict all future states and outputs.

Indistinguishability of Parameters. In the honest generation, g and h are sampled independently and uniformly from QR_p. In the NSA generation, g is sampled rndly from QR_p, and then h is computed as g^c, where c is sampled uniformly from $\mathbb{Z}_q$. Now because QR_p is a cyclic group of prime order q, for any fixed $g \neq 1$, the map $c \mapsto g^c$ will be a bijection from $\mathbb{Z}_q$ to QR_p. Therefore, when c is uniform in $\mathbb{Z}_q$, $h = g^c$ will be uniformly distributed over QR_p, and is therefore statistically independent of g.

Consequently, the joint distribution of (g, h) produced by the NSA generator is *the same* to the joint distribution produced by the honest generator. The statistical distance between the two distributions is just 0, making them indistinguishable from one another. An adv. will not be able to detect the presence of the backdoor without first even knowing that sk exists.